JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Finite Dimensional Vector Spaces (2015–2021)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
28	44 M	84 Min	MCQ • MSQ • NAT (2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	12	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	5	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	11	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q60 • ID: JAM_2021_Q60) **NAT** +2 M Neg: Nil (0)

Q. 60 Let $A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$. Consider the linear map T_A from the real vector space $M_4(\mathbb{R})$ to itself defined by $T_A(X) = AX - XA$, for all $X \in M_4(\mathbb{R})$. The dimension of the range of T_A is ______.

Question 2 (JAM 2021 • Q46 • ID: JAM_2021_Q46) **NAT** +1 M Neg: Nil (0)

Q. 46 Let V be the real vector space of all continuous functions $f : [0, 2] \to \mathbb{R}$ such that the restriction of f to the interval $[0, 1]$ is a polynomial of degree less than or equal to 2, the restriction of f to the interval $[1, 2]$ is a polynomial of degree less than or equal to 3 and $f(0) = 0$. Then the dimension of V is equal to ______.

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Question 3 (JAM 2021 • Q39 • ID: JAM_2021_Q39) **MSQ** +2 M Neg: Nil (0)

Q. 39 Let V be a finite dimensional vector space and $T : V \to V$ be a linear transformation. Let $\mathcal{R}(T)$ denote the range of T and $\mathcal{N}(T)$ denote the null space $\{v \in V : Tv = 0\}$ of T. If $\text{rank}(T) = \text{rank}(T^2)$, then which of the following is/are necessarily true?

(A) $\mathcal{N}(T) = \mathcal{N}(T^2)$.

(B) $\mathcal{R}(T) = \mathcal{R}(T^2)$.

(C) $\mathcal{N}(T) \cap \mathcal{R}(T) = \{0\}$.

(D) $\mathcal{N}(T) = \{0\}$.

Question 4 (JAM 2021 • Q29 • ID: JAM_2021_Q29) MCQ +2 M Neg: −0.67

Q. 29 Let $n \geq 2$ be an integer. Let $A : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ be the linear transformation defined by

$$A(z_1, z_2, \ldots, z_n) = (z_n, z_1, z_2, \ldots, z_{n-1}).$$

Which one of the following statements is true for every $n \geq 2$?

(A) A is nilpotent.
(B) All eigenvalues of A are of modulus 1.
(C) Every eigenvalue of A is either 0 or 1.
(D) A is singular.

Question 5 (JAM 2021 • Q11 • ID: JAM_2021_Q11) MCQ +2 M Neg: −0.67

Q. 11 – Q. 30 carry two marks each.

Q. 11 Let $M_n(\mathbb{R})$ be the real vector space of all $n \times n$ matrices with real entries, $n \geq 2$. Let $A \in M_n(\mathbb{R})$. Consider the subspace W of $M_n(\mathbb{R})$ spanned by $\{I_n, A, A^2, \ldots\}$. Then the dimension of W over $\mathbb{R}$ is necessarily

(A) ∞.
(B) n^2.
(C) n.
(D) at most n.

Question 6 (JAM 2021 • Q9 • ID: JAM_2021_Q9) MCQ +1 M Neg: −0.33

Q. 9 For an integer $k \geq 0$, let P_k denote the vector space of all real polynomials in one variable of degree less than or equal to k. Define a linear transformation $T : P_2 \longrightarrow P_3$ by

$$Tf(x) = f''(x) + xf(x).$$

Which one of the following polynomials is not in the range of T?

(A) $x + x^2$
(B) $x^2 + x^3 + 2$
(C) $x + x^3 + 2$
(D) $x + 1$

Question 7 (JAM 2020 • Q59 • ID: JAM_2020_Q59) **NAT** **+2 M** **Neg: Nil (0)**

Q. 59 Let $T : \mathbb{R}^7 \to \mathbb{R}^7$ be a linear transformation with Nullity$(T) = 2$. Then, the minimum possible value for Rank(T^2) is ________.

Question 8 (JAM 2020 • Q49 • ID: JAM_2020_Q49) **NAT** **+1 M** **Neg: Nil (0)**

Q. 49 Consider the real vector space $P_{2020} = \{\sum_{i=0}^{n} a_i x^i : a_i \in \mathbb{R} \text{ and } 0 \le n \le 2020\}$. Let W be the subspace given by

$$W = \left\{ \sum_{i=0}^{n} a_i x^i \in P_{2020} : a_i = 0 \text{ for all odd } i \right\}.$$

Then, the dimension of W is ________.

Question 9 (JAM 2020 • Q34 • ID: JAM_2020_Q34) **MSQ** **+2 M** **Neg: Nil (0)**

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Q. 34 Let V be a non-zero vector space over a field F. Let $S \subset V$ be a non-empty set. Consider the following properties of S:

(I) For any vector space W over F, any map $f : S \to W$ extends to a linear map from V to W.

(II) For any vector space W over F and any two linear maps $f, g : V \to W$ satisfying $f(s) = g(s)$ for all $s \in S$, we have $f(v) = g(v)$ for all $v \in V$.

(III) S is linearly independent.

(IV) The span of S is V.

Which of the following statement(s) is /are true?

(A) (I) implies (IV)

(B) (I) implies (III)

(C) (II) implies (III)

(D) (II) implies (IV)

Question 10 (JAM 2020 • Q8 • ID: JAM_2020_Q8) MCQ +1 M Neg: −0.33

Q. 8 Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation given by $T(x, y) = (-x, y)$. Then

(A) $T^{2k} = T$ for all $k \geq 1$

(B) $T^{2k+1} = -T$ for all $k \geq 1$

(C) the range of T^2 is a proper subspace of the range of T

(D) the range of T^2 is equal to the range of T

Question 11 (JAM 2019 • Q42 • ID: JAM_2019_Q42) NAT +1 M Neg: Nil (0)

Q.42 Let W_1 and W_2 be subspaces of the real vector space $\mathbb{R}^{100}$ defined by

$$W_1 = \{ (x_1, x_2, \dots, x_{100}) : x_i = 0 \text{ if } i \text{ is divisible by } 4\},$$

$$W_2 = \{ (x_1, x_2, \dots, x_{100}) : x_i = 0 \text{ if } i \text{ is divisible by } 5\}.$$

Then the dimension of $W_1 \cap W_2$ is _____

Question 12 (JAM 2019 • Q33 • ID: JAM_2019_Q33) MSQ +2 M Neg: Nil (0)

Q.33 Let S and T be linear transformations from a finite dimensional vector space V to itself such that $S\big(T(v)\big) = 0$ for all $v \in V$. Then

(A) $\text{rank}(T) \geq \text{nullity}(S)$
(B) $\text{rank}(S) \geq \text{nullity}(T)$
(C) $\text{rank}(T) \leq \text{nullity}(S)$
(D) $\text{rank}(S) \leq \text{nullity}(T)$

Question 13 (JAM 2019 • Q2 • ID: JAM_2019_Q2) MCQ +1 M Neg: −0.33

Q.2 Let $T \in M_{m\times n}(\mathbb{R})$. Let V be the subspace of $M_{n\times p}(\mathbb{R})$ defined by

$$V = \{X \in M_{n\times p}(\mathbb{R}) : TX = 0\}.$$

Then the dimension of V is

(A) $pn - \text{rank}(T)$
(B) $mn - p\,\text{rank}(T)$
(C) $p(m - \text{rank}(T))$
(D) $p(n - \text{rank}(T))$

Question 14 (JAM 2018 • Q57 • ID: JAM_2018_Q57) **NAT** **+2 M** **Neg: Nil (0)**

Q.57 Suppose $Q \in M_{3\times3}(\mathbb{R})$ is a matrix of rank 2. Let $T: M_{3\times3}(\mathbb{R}) \to M_{3\times3}(\mathbb{R})$ be the linear transformation defined by $T(P) = QP$. Then the rank of T is ______________

Question 15 (JAM 2018 • Q49 • ID: JAM_2018_Q49) **NAT** **+1 M** **Neg: Nil (0)**

Q.49 Let W_1 be the real vector space of all 5×2 matrices such that the sum of the entries in each row is zero. Let W_2 be the real vector space of all 5×2 matrices such that the sum of the entries in each column is zero. Then the dimension of the space $W_1 \cap W_2$ is _____________

Question 16 (JAM 2018 • Q21 • ID: JAM_2018_Q21) **MCQ** **+2 M** **Neg: −0.67**

Q.21 Let U, V and W be finite dimensional real vector spaces, $T: U \to V$, $S: V \to W$ and $P: W \to U$ be linear transformations. If range (ST) = nullspace (P), nullspace (ST) = range (P) and rank (T) = rank (S), then which one of the following is TRUE?

(A) nullity of T = nullity of S

(B) dimension of $U \neq$ dimension of W

(C) If dimension of $V = 3$, dimension of $U = 4$, then P is not identically zero

(D) If dimension of $V = 4$, dimension of $U = 3$ and T is one-one, then P is identically zero

Question 17 (JAM 2018 • Q9 • ID: JAM_2018_Q9) **MCQ** **+1 M** **Neg: −0.33**

Q.9 Consider the vector space V over $\mathbb{R}$ of polynomial functions of degree less than or equal to 3 defined on $\mathbb{R}$. Let $T : V \to V$ be defined by $(Tf)(x) = f(x) - xf'(x)$. Then the rank of T is

(A) 1 (B) 2 (C) 3 (D) 4

Question 18 (JAM 2018 • Q3 • ID: JAM_2018_Q3) MCQ +1 M Neg: −0.33

Q.3 If $\{v_1, v_2, v_3\}$ is a linearly independent set of vectors in a vector space over $\mathbb{R}$, then which one of the following sets is also linearly independent?

(A) $\{v_1 + v_2 - v_3,\ 2v_1 + v_2 + 3v_3,\ 5v_1 + 4v_2\}$

(B) $\{v_1 - v_2,\ v_2 - v_3,\ v_3 - v_1\}$

(C) $\{v_1 + v_2 - v_3,\ \ v_2 + v_3 - v_1,\ \ v_3 + v_1 - v_2,\ \ v_1 + v_2 + v_3\}$

(D) $\{v_1 + v_2,\ \ v_2 + 2v_3,\ \ v_3 + 3v_1\}$

Question 19 (JAM 2017 • Q46 • ID: JAM_2017_Q46) NAT +1 M Neg: Nil (0)

Q.46 Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Let M be the matrix whose columns are $v_1, v_2, 2v_1 - v_2, v_1 + 2v_2$ in that order. Then the number of linearly independent solutions of the homogeneous system of linear equations $Mx = 0$ is __________

Question 20 (JAM 2017 • Q22 • ID: JAM_2017_Q22) MCQ +2 M Neg: −0.67

Q.22 Let $\mathcal{P}_3$ denote the real vector space of all polynomials with real coefficients of degree at most 3. Consider the map $T: \mathcal{P}_3 \to \mathcal{P}_3$ given by $T(p(x)) = p''(x) + p(x)$. Then

(A) T is neither one-one nor onto

(B) T is both one-one and onto

(C) T is one-one but not onto

(D) T is onto but not one-one

Question 21 (JAM 2016 • Q59 • ID: JAM_2016_Q59) NAT +2 M Neg: Nil (0)

Q.59 Let $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 5 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 1 \end{bmatrix}$, $N(A)$ the null space of A and $R(B)$ the range space of B.
Then the dimension of $N(A) \cap R(B)$ over $\mathbb{R}$ is ___________

Question 22 (JAM 2016 • Q39 • ID: JAM_2016_Q39) MSQ +2 M Neg: Nil (0)

Q.39 Consider the set $V = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; \alpha x + \beta y + z = \gamma,\ \alpha, \beta, \gamma \in \mathbb{R} \right\}$. For which of the following choice(s) the set V becomes a two dimensional subspace of $\mathbb{R}^3$ over $\mathbb{R}$?

(A) $\alpha = 0,\ \beta = 1,\ \gamma = 0$
(B) $\alpha = 0,\ \beta = 1,\ \gamma = 1$
(C) $\alpha = 1,\ \beta = 0,\ \gamma = 0$
(D) $\alpha = 1,\ \beta = 1,\ \gamma = 0$

Question 23 (JAM 2016 • Q2 • ID: JAM_2016_Q2) MCQ +1 M Neg: −0.33

Q.2 Let P be the vector space (over $\mathbb{R}$) of all polynomials of degree ≤ 3 with real coefficients. Consider the linear transformation $T: P \to P$ defined by

$$T(a_0 + a_1 x + a_2 x^2 + a_3 x^3) = a_3 + a_2 x + a_1 x^2 + a_0 x^3.$$

Then the matrix representation M of T with respect to the ordered basis $\{1, x, x^2, x^3\}$ satisfies

(A) $M^2 + I_4 = 0$ (B) $M^2 - I_4 = 0$
(C) $M - I_4 = 0$ (D) $M + I_4 = 0$

Question 24 (JAM 2015 • Q54 • ID: JAM_2015_Q54) **NAT** **+2 M** **Neg: Nil (0)**

Q.14 Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. Let V be a subspace of $M_2(\mathbb{R})$ defined by

$$V = \left\{ A \in M_2(\mathbb{R}) : A \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} A \right\}$$

Then the dimension of V is ______________

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Question 25 (JAM 2015 • Q48 • ID: JAM_2015_Q48) **NAT** **+1 M** **Neg: Nil (0)**

Q.8

If the set $\left\{ \begin{bmatrix} x & -x \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ x & x \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} \right\}$ is linearly dependent in the vector space of all 2×2 matrices with real entries, then x is equal to ______________

Question 26 (JAM 2015 • Q35 • ID: JAM_2015_Q35) **MSQ** **+2 M** **Neg: Nil (0)**

Q.5

Let V be the set of 2×2 matrices $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ with complex entries such that $a_{11} + a_{22} = 0$. Let W be the set of matrices in V with $a_{12} + \overline{a_{21}} = 0$. Then, under usual matrix addition and scalar multiplication, which of the following is (are) true?

(A) V is a vector space over $\mathbb{C}$
(B) W is a vector space over $\mathbb{C}$
(C) V is a vector space over $\mathbb{R}$
(D) W is a vector space over $\mathbb{R}$

Question 27 (JAM 2015 • Q19 • ID: JAM_2015_Q19) MCQ +2 M Neg: −0.67

Q.19

Let $B_1 = \{(1,2),\ (2,-1)\}$ and $B_2 = \{(1,0),\ (0,1)\}$ be ordered bases of $\mathbb{R}^2$. If $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation such that $[T]_{B_1,B_2}$, the matrix of T with respect to B_1 and B_2, is $\begin{bmatrix} 4 & 3 \\ 2 & -4 \end{bmatrix}$, then $T(5,5)$ is equal to

(A) $(-9,\ 8)$ (B) $(9,\ 8)$ (C) $(-15,\ -2)$ (D) $(15,\ 2)$

Question 28 (JAM 2015 • Q18 • ID: JAM_2015_Q18) MCQ +2 M Neg: −0.67

Q.18 Let $P_2(\mathbb{R})$ be the vector space of polynomials in x of degree at most 2 with real coefficients. Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. If a linear transformation $T: P_2(\mathbb{R}) \to M_2(\mathbb{R})$ is defined as

$$T(f) = \begin{bmatrix} f(0) - f(2) & 0 \\ 0 & f(1) \end{bmatrix}$$

then

(A) T is one-one but not onto
(B) T is onto but not one-one
(C) Range (T) = span $\left\{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix}\right\}$
(D) Null (T) = span $\{x^2 - 2x,\ 1 - x\}$

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OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Finite Dimensional Vector Spaces (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	**8**
2	NAT	+1	**5**
3	MSQ	+2	**A, B, C**
4	MCQ	+2	**B**
5	MCQ	+2	**D**
6	MCQ	+1	**D**
7	NAT	+2	**3 to 3**
8	NAT	+1	**1011 to 1011**
9	MSQ	+2	**B;D**
10	MCQ	+1	**D**
11	NAT	+1	**60 to 60**
12	MSQ	+2	**C, D**
13	MCQ	+1	**D**
14	NAT	+2	**6 to 6**
15	NAT	+1	**4 to 4**
16	MCQ	+2	**C**
17	MCQ	+1	**C**
18	MCQ	+1	**D**
19	NAT	+1	**1.9 to 2.1**
20	MCQ	+2	**B**
21	NAT	+2	**1.0 to 1.0**
22	MSQ	+2	**A;C;D**
23	MCQ	+1	**B**
24	NAT	+2	**2**
25	NAT	+1	**-1**
26	MSQ	+2	**A;C;D**
27	MCQ	+2	**D**
28	MCQ	+2	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.